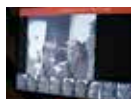




Older or younger doctors?

Study by Harvard researchers found that the older the doctor is, the higher the patient mortality rate. <http://digit.in/OldDocs>



Facial recognition in school

A B-school in Paris is using facial recognition to find out when students aren't paying attention in class. <http://digit.in/schoolface>

The Origins of Indians

Are we made up of travellers from across the planet?

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India is the seventh-largest country in the world (in terms of area) and is also synonymous with being the world's biggest democracy. The world now knows us for those facts and also the progress we've made in the technology space. But "India" isn't a homogenous entity. Even a casual observer won't miss the diversity of cultures, sects, beliefs, languages, cuisines and religions found in this land. The cultural and food habits aside, the physical appearance of Indians also differs from region to region.

Some of the more recent migration waves, which have been well documented (by recent we mean about 1500 years) do explain part of this diversity. But genetics can go far beyond that minuscule time span. Let's take a trip down the timeline. Our first stop is about 4200 thousand years ago.

ANCESTRAL COMPONENTS IN CONTEMPORARY INDIAN POPULATION

For a long time, the Indian gene pool was divided into two groups: the Ancestral North Indians (ANI) and the Ancestral South Indians (ASI). Physically, several differences exist between the North Indians – who generally have a fairer complexion vis-à-vis their darker South Indian counterparts,

Scientists believe that the ANI are genetically close to Middle Eastern-



Where Does The DNA In Modern Indians Originate

ers, Central Asians and Europeans. They also posit that the ASI were distinct from ANI and not closely related to any other groups outside the subcontinent. The ASI are the earliest settlers in India, possibly arriving on the southern exit wave from Africa. However between 4,200 and 1,900 years ago, the groups began intermingling. This migration was prompted by a search for greener pastures. Further, it is believed that the migration of the West Asian people together with the spread of agricultural methods around 4,600 BP (Before Present or Before 1950) led to the mass movement.

PALAEOLITHIC MIGRATION THEORIES

Now let us go further back in time. When did the earliest successful migration of modern humans out of Africa occur? Some theories suggest that the migration happened either pre- or post-Toba. For the unfamiliar, Toba was a catastrophic volcanic eruption that occurred 69,000 to 77,000

years ago at the site of modern-day Lake Toba. This 30km-wide lake is in Indonesia and is in the vicinity of a supervolcano. The Toba eruption – the biggest in 25 million years – threw up so much ash into the atmosphere, it is believed to brought many species to the brink of extinction - including humans. However, it might have prompted a new wave of global migration too.

In 2007, palaeontologists unearthed several stone tools below the layers of ash disposed in India at Jwalapuram, Andhra Pradesh, which point to a pre-Toba dispersal. These stone tools were supposedly made by ancestors of present-day Indians. The tools seemed to be from an earlier era and it is believed that the ash from the Toba volcano eruption impacted the area, which was previously thriving.

Prior to this discovery, a school of thought existed that people migrated from Africa sometime around 60,000 years to 70,000 years ago, which was a few thousand years post Toba.

Another hypothesis is that all people who existed in India before the Toba eruption went extinct shortly after the eruption blanketed the Indian subcontinent in ash. These conditions may have been too hard for the population to withstand, which is why no one survived. If true, this would indicate that the DNA in modern Indians has no link to these early settlers who went extinct around the Toba activity.

In 2009, Scientists excavated the Toba ash in India to find tools belonging both to the period before the eruption and after, indicating that at least some small groups managed to survive the catastrophe and lived on. No human fossils were discovered, however.

After the Toba eruption, around 150 to 1,000 people from Africa crossed the Red Sea and found their way to India. These may essentially be the first settlers from Africa in India.

ORIGINS OF DRAVIDIANS

The circumstances of the advent of Dravidian speakers in India have been an enigma. Vague linguistic and cul-

Image credit: Earl McGehee | Flickr